

Supplementary Material S1
for “Pareto-Safe Pair Reassignment for LSCBO-Based Multi-Cost Cloud Task
Scheduling”

S1. Purpose of This Supplement

This supplementary note accompanies the CEC2017 benchmark section in the main manuscript. Its purpose is to report the full benchmark coverage and the aggregate statistical protocol used in the paper.

S2. Full Benchmark Coverage Used in Section 5.1

The auxiliary benchmark section in the main manuscript evaluates 29 functions from the CEC2017 single-objective suite. Function F2 is deprecated in the official problem definition; all remaining functions used in the manuscript are listed below.

Table 1: Full function coverage for the CEC2017 section in the manuscript

Function ID	Landscape Family	Role in the manuscript
F1	Unimodal	Included in the full 29-function statistical analysis.
F3	Unimodal	Included in the full 29-function statistical analysis.
F4	Simple multimodal	Included in the full 29-function statistical analysis.
F5	Simple multimodal	Included in the full 29-function statistical analysis.
F6	Simple multimodal	Included in the full 29-function statistical analysis.
F7	Simple multimodal	Included in the full 29-function statistical analysis.
F8	Simple multimodal	Included in the full 29-function statistical analysis.
F9	Simple multimodal	Included in the full 29-function statistical analysis.
F10	Simple multimodal	Included in the full 29-function statistical analysis.
F11	Hybrid	Included in the full 29-function statistical analysis.
F12	Hybrid	Included in the full 29-function statistical analysis.
F13	Hybrid	Included in the full 29-function statistical analysis.
F14	Hybrid	Included in the full 29-function statistical analysis.
F15	Hybrid	Included in the full 29-function statistical analysis.

Table 2: Full function coverage for the CEC2017 section in the manuscript (continued)

Function ID	Landscape Family	Role in the manuscript
F16	Hybrid	Included in the full 29-function statistical analysis.
F17	Hybrid	Included in the full 29-function statistical analysis.
F18	Hybrid	Included in the full 29-function statistical analysis.
F19	Hybrid	Included in the full 29-function statistical analysis.
F20	Hybrid	Included in the full 29-function statistical analysis.
F21	Composition	Included in the full 29-function statistical analysis.
F22	Composition	Included in the full 29-function statistical analysis.
F23	Composition	Included in the full 29-function statistical analysis.
F24	Composition	Included in the full 29-function statistical analysis.
F25	Composition	Included in the full 29-function statistical analysis.
F26	Composition	Included in the full 29-function statistical analysis.
F27	Composition	Included in the full 29-function statistical analysis.
F28	Composition	Included in the full 29-function statistical analysis.
F29	Composition	Included in the full 29-function statistical analysis.
F30	Composition	Included in the full 29-function statistical analysis.

S3. Aggregate Statistical Outputs Used in the Main Manuscript

Section 5.1 of the main manuscript reports rank-based statistics computed from 30 paired runs on the full 29-function set. The comparison contains nine algorithms and 7,830 completed runs. Pairwise post-hoc tests are performed on the 29 per-function mean errors rather than by pooling all 870 function–run observations, so the supplementary summary follows the same cross-function protocol as the main text.

Table 3: Friedman ranking over the full 29-function benchmark set

Algorithm	Average Rank	Relative Position
LSCBO	1.690	1st
SA	2.655	2nd
PSO	3.345	3rd
HLBO	4.621	4th
GWCA	4.793	5th
CBO	5.034	6th
GWO	6.862	7th
AOA	7.448	8th
WOA	8.552	9th

Table 4: Holm-corrected cross-function Wilcoxon summary for LSCBO against the other eight algorithms over 29 per-function mean errors

Comparison	Wins–losses	Holm-adjusted p-value	Interpretation
LSCBO vs CBO	29–0	2.16×10^{-5}	LSCBO lower
LSCBO vs PSO	22–7	0.00108	LSCBO lower
LSCBO vs GWO	29–0	2.16×10^{-5}	LSCBO lower
LSCBO vs WOA	29–0	2.16×10^{-5}	LSCBO lower
LSCBO vs SA	19–10	0.0661	Not distinguished
LSCBO vs AOA	29–0	2.16×10^{-5}	LSCBO lower
LSCBO vs HLBO	28–1	8.19×10^{-5}	LSCBO lower
LSCBO vs GWCA	27–2	6.74×10^{-5}	LSCBO lower

The Nemenyi critical difference is approximately 2.23. The rank gaps from LSCBO to SA (0.97) and PSO (1.66) are below this threshold. The pairwise Wilcoxon–Holm and all-pairs Nemenyi procedures answer different post-hoc questions; the main manuscript therefore reports the exact pairwise outcomes and avoids the claim that LSCBO is significantly superior to every comparator.

S4. Availability Note for the Benchmark Materials

The reproducibility package contains all 7,830 per-run rows, per-function descriptive statistics, convergence checkpoints, the complete rank table, paired statistical outputs, the locked parameter record, and the official-function provenance information. This supplement is limited to benchmark coverage and statistical protocol; the analytical scheduling study and its 12-row CloudSim Plus smoke replay are reported separately in Section 5.2 of the main manuscript.